

Table 1 – Minimum width of joint and maximum gap for enclosures of groups I, IIA and IIB

Type of joint		Minimum width of joint L mm	Maximum gap mm											
			For a volume cm^3 $V \leq 100$			For a volume cm^3 $100 < V \leq 500$			For a volume cm^3 $500 < V \leq 2\,000$			For a volume cm^3 $V > 2\,000$		
			I	IIA	IIB	I	IIA	IIB	I	IIA	IIB	I	IIA	IIB
Flanged, cylindrical or spigot joints		6	0,30	0,30	0,20	—	—	—	—	—	—	—	—	—
		9,5	0,35	0,30	0,20	0,35	0,30	0,20	0,08	0,08	0,08	—	—	—
		12,5	0,40	0,30	0,20	0,40	0,30	0,20	0,40	0,30	0,20	0,40	0,20	0,15
		25	0,50	0,40	0,20	0,50	0,40	0,20	0,50	0,40	0,20	0,50	0,40	0,20
Cylindrical joints for shaft glands of rotating electrical machines with:	Sleeve bearings	6	0,30	0,30	0,20	—	—	—	—	—	—	—	—	—
		9,5	0,35	0,30	0,20	0,35	0,30	0,20	—	—	—	—	—	—
		12,5	0,40	0,35	0,25	0,40	0,30	0,20	0,40	0,30	0,20	0,40	0,20	—
		25	0,50	0,40	0,30	0,50	0,40	0,25	0,50	0,40	0,25	0,50	0,40	0,20
		40	0,60	0,50	0,40	0,60	0,50	0,30	0,60	0,50	0,30	0,60	0,50	0,25
	Rolling-element bearings	6	0,45	0,45	0,30	—	—	—	—	—	—	—	—	—
		9,5	0,50	0,45	0,35	0,50	0,40	0,25	—	—	—	—	—	—
		12,5	0,60	0,50	0,40	0,60	0,45	0,30	0,60	0,45	0,30	0,60	0,30	0,20
		25	0,75	0,60	0,45	0,75	0,60	0,40	0,75	0,60	0,40	0,75	0,60	0,30
		40	0,80	0,75	0,60	0,80	0,75	0,45	0,80	0,75	0,45	0,80	0,75	0,40

NOTE Constructional values rounded according to ISO 31-0 should be taken when determining the maximum gap.

Table 2 – Minimum width of joint and maximum gap for group IIC enclosures

Type of joint		Minimum width of joint L mm	Maximum gap mm			
			For a volume cm^3 $V \leq 100$	For a volume cm^3 $100 < V \leq 500$	For a volume cm^3 $500 < V \leq 2\,000$	For a volume cm^3 $V > 2\,000$
Flanged joints ^a		6	0,10	–	–	–
		9,5	0,10	0,10	–	–
		15,8	0,10	0,10	0,04	–
		25	0,10	0,10	0,04	0,04
Spigot joints (Figure 2a)	$c \geq 6 \text{ mm}$	12,5	0,15	0,15	0,15	–
	$d \geq 0,5 L$	25	0,18 ^b	0,18 ^b	0,18 ^b	0,18 ^b
	$L = c + d$	40	0,20 ^c	0,20 ^c	0,20 ^c	0,20 ^c
	$f \leq 1 \text{ mm}$					
Cylindrical joints Spigot joints (Figure 2b)		6	0,10	–	–	–
		9,5	0,10	0,10	–	–
		12,5	0,15	0,15	0,15	–
		25	0,15	0,15	0,15	0,15
Cylindrical joints for shaft glands of rotating electrical machines with rolling element bearings		40	0,20	0,20	0,20	0,20
		6	0,15	–	–	–
		9,5	0,15	0,15	–	–
		12,5	0,25	0,25	0,25	–
		25	0,25	0,25	0,25	0,25
		40	0,30	0,30	0,30	0,30
^a Flanged joints are permitted for explosive mixtures of acetylene and air only in accordance with 5.2.7. ^b Maximum gap of cylindrical part increased to 0,20 mm if $f < 0,5 \text{ mm}$. ^c Maximum gap of cylindrical part increased to 0,25 mm if $f < 0,5 \text{ mm}$.						
NOTE The constructional values rounded according to ISO 31-0 should be taken when determining the maximum gap.						

5.3 Threaded joints

Threaded joints shall comply with the requirements given in Tables 3 or 4.

Table 3 – Cylindrical threaded joints

Pitch	$\geq 0,7 \text{ mm}^a$
Thread form and quality of fit	Medium or fine tolerance quality according to ISO 965-1 and ISO 965-3 ^b
Threads engaged	≥ 5
Depth of engagement	
Volume $\leq 100 \text{ cm}^3$	$\geq 5 \text{ mm}$
Volume $> 100 \text{ cm}^3$	$\geq 8 \text{ mm}$
^a Where the pitch exceeds 2 mm, special manufacturing precautions may be necessary (for example, more threads engaged) to ensure that the electrical equipment can pass the test for non-transmission of an internal ignition which is prescribed in 15.2. ^b Cylindrical threaded joints which do not conform with ISO 965-3 in respect of thread form or quality of fit, are not permitted. If the test for non-transmission of an internal ignition, prescribed in 15.2, is passed, when the width of the thread is as specified in Table 3.	